Büchi Automata and ω-regular Languages

A **Büchi Automaton** is a type of NFA with no ε-transitions that accepts inputs in Σ^ω, i.e. infinite length inputs.

Σ^ω represents any infinite word over alphabet Σ
Σ^* represents any finite word over alphabet Σ

A Büchi Automaton accepts an input string $w_1 w_2 \ldots \in \Sigma^\omega$ if there exists a path through the machine (starting at a starting state and following the transition function at each step) such that there is an infinite number of accepting states in this path.
Since FSMs have a finite number of states, this means that at least one state in the BA must be visited infinitely often.

A language is **ω-regular** iff it is recognised by some BA.

Operations:

Union: $A \cup B$, A and B ω-regular

Concatenation: AB, A is regular, B is ω-regular

Two ω-regular languages cannot be concatenated as they deal with infinite length words.
Similarly, A cannot be ω-regular with B regular, since we never get to B.

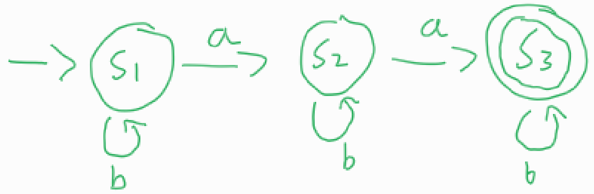
ω: A^ω – an infinite sequence of words from A, where A is regular and does not contain the empty word.

All these operations produce an ω-regular language.

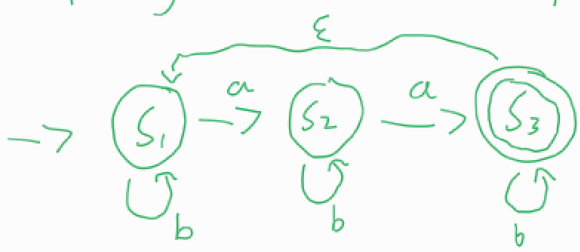
Example

L is a regular language recognised by the following NFA:

$L = b^*ab^*ab^*$

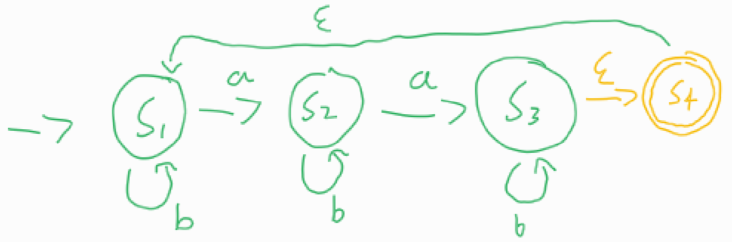


The following is not a valid Büchi for L^ω:

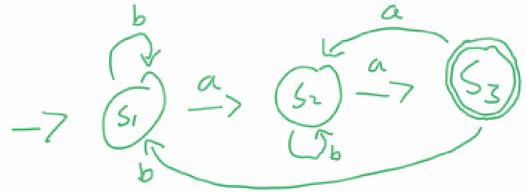


1. Büchi automata cannot have ε-transitions.
2. This automata accepts the string aab^ω which is not in L^ω as it is not an infinite sequence of words in L, but visits accepting state S_3 infinitely often.

Fix for (2) - The BA is forced to go back to the start after reaching the accept state.



Fix for (1) and (2):



Note: the initial invalid Büchi accepts Lim L.

Lim L is the language $\{a \in \Sigma^\omega \mid a$ has infinitely many prefixes in $L\}$

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this means that at as you read the infinite sequence, there are infinitely many points such that the string read so far is a valid string in L.

All possible infinite strings where this is the case must be in Lim L.

$L^\omega \subseteq \text{Lim} L$, but Lim L also includes aab^ω etc → doesn't require infinitely many words from L.

An ω-regular language is the limit of some regular language iff some <u>deterministic</u> BA recognises it.

<u>Example</u>

Consider a regular language $L = \{0,1\}^*0$, i.e the set of all binary strings ending in 0.

Which of these ω-regular languages is the limit of L:

- $A: (1^*0)^\omega$
- $B = \{0,1\}^*0^\omega$

The limit of L is the set of <u>all</u> infinite words with infinitely many prefixes ending in 0.

The set B obviously contains infinite words with infinitely many prefixes ending in 0. But is it all of them?

Counterexample: $(10)^\omega$ is not in B but has infinitely many prefixes ending in 0.

Also $(110)^\omega$, $(1110)^\omega$, ..., $(1^*0)^\omega$

$\hookrightarrow$ This is A.

$\therefore$ A is the limit of L, not B. B is a subset of the limit.